



Scratch

Question Paper *Phiếu câu hỏi*

Time allowed: 45 minutes

Thời gian thi: 45 phút

Instructions to Contestants: *(Hướng dẫn thi)*

1. Each contestant should have ONE Question-Answer Book which CANNOT be taken away.
Mỗi thí sinh được cấp MỘT bộ Phiếu câu hỏi - Phiếu trả lời. Thí sinh KHÔNG ĐƯỢC mang tài liệu ra ngoài phòng thi.
2. There are a total of 16 questions in this Question-Answer Book. Two points for a correct answer. No points for blank answers. ONE penalty point will be deducted for an incorrect answer.
Đề thi gồm 16 câu hỏi. Mỗi câu trả lời đúng được 2 điểm. Câu không trả lời không bị trừ điểm. Mỗi câu trả lời sai sẽ bị trừ MỘT điểm.
3. All answers should be written on ANSWER SHEET.
Tất cả đáp án phải được điền vào PHIẾU TRẢ LỜI.
4. During the contest, NO calculators can be used for Scratch and Blockly but calculators can be used for Python.
Thí sinh thi ngôn ngữ Scratch và Blockly KHÔNG được dùng máy tính cầm tay. Thí sinh thi ngôn ngữ Python được dùng máy tính cầm tay.
5. This Question-Answer Book will be collected at the end of the contest.
Bộ Phiếu câu hỏi – Phiếu trả lời sẽ được thu lại sau khi thí sinh hoàn thành bài thi.
6. DO NOT turn over this Question-Answer Book without approval of the examiner.
Thí sinh KHÔNG tự ý xem đề khi chưa có hiệu lệnh của giám thị.

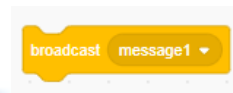
Otherwise, contestant may be DISQUALIFIED.

Nếu thí sinh vi phạm một trong các điều trên, BTC có quyền hủy kết quả thi của thí sinh đó.

1. Scratch is developed by whom?

- A. Mitchel Resnick
- B. Tim Berners Lee
- C. Herman Hollerith
- D. Microsoft
- E. None of the above

2. Which group of blocks includes the "Broadcast" block?



- A. Motion B. Events C. Control D. Looks E. Sound

3. I go to add a script to my character and find that my blocks have disappeared. Why is this?

- A. Scratch Malfunctioned
- B. I have stage selected
- C. I have exceeded the block limit
- D. All of the above
- E. None of the above

4. Which block will give the sprite a location to start from when placed at the start of your code?

- A. Move B. Switch Costume C. Go To D. Show E. Hide

5. Where do you find the background images to insert in Scratch?

- A. Costumes B. Gallery C. Script D. Backdrops E. Sprite

6. A computer follows step-wise instructions to complete any task, which is known as _____?

- A. Program or Script B. Gallery C. Sprite D. Scratch E. None of the above

7. How do you create a loop in scratch?

- A. Use a cat sprite
- B. Use a repeat block
- C. Snap a block
- D. Use a backdrop
- E. None of the above

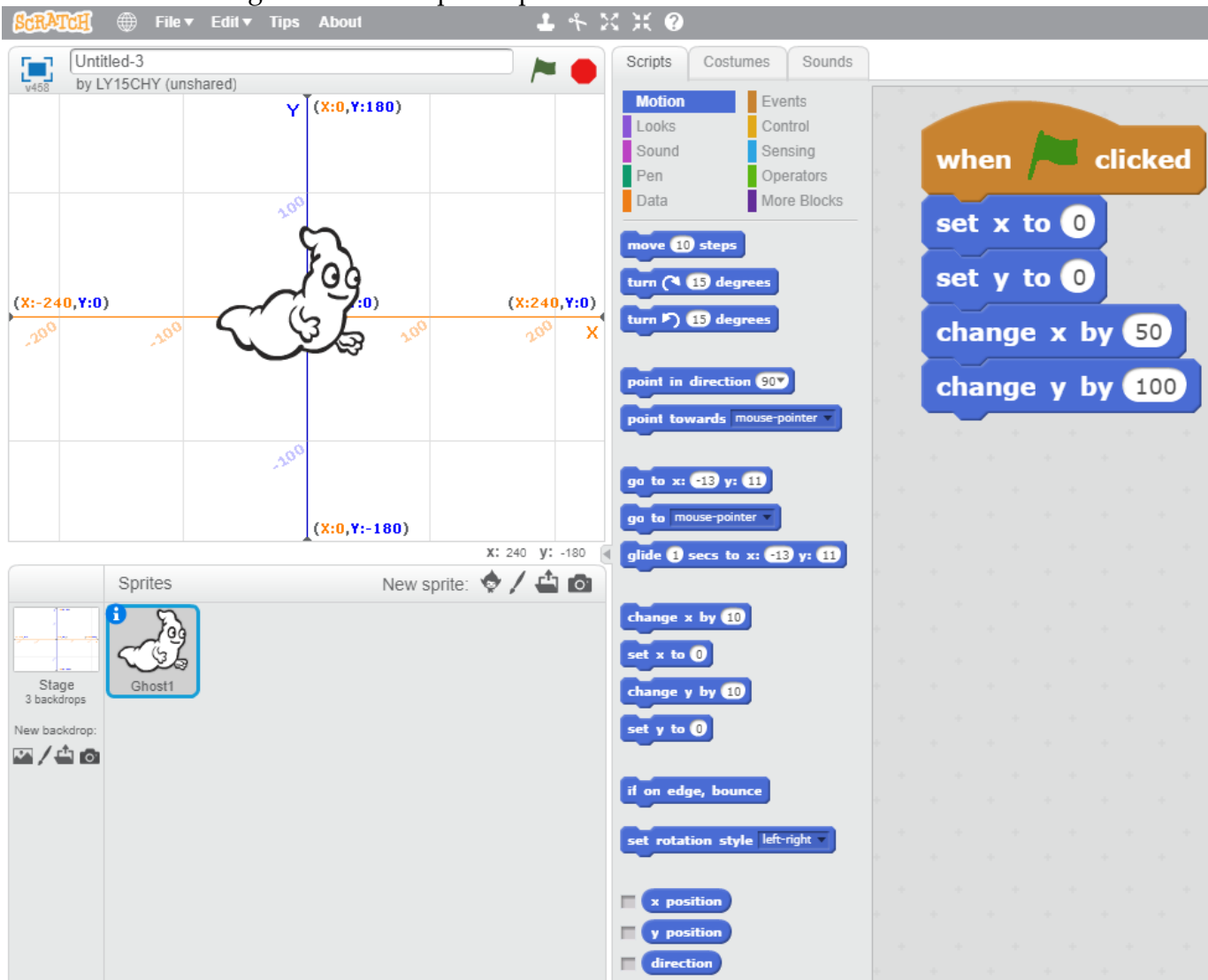
8. What is the value of solution?

- A. 5
- B. 23
- C. 19+4
- D. Solution
- E. None of the above



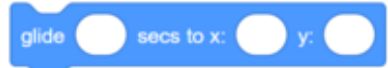
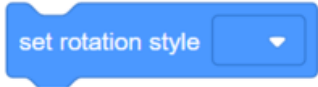
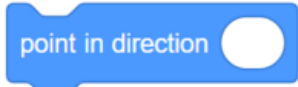
All answers should be written on the ANSWER SHEET.

Refer to the following scene and script for question 9 and 10.

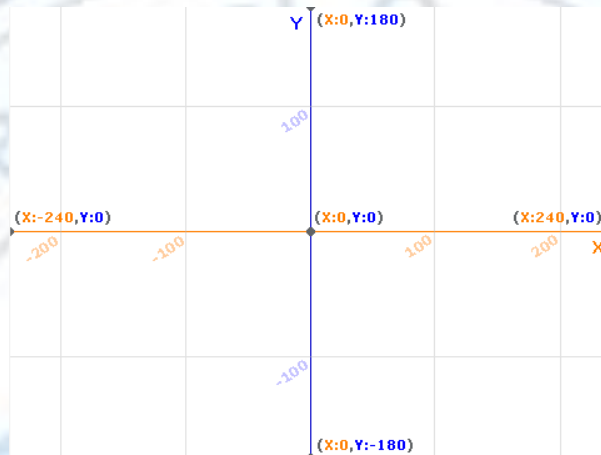


9. Looking at the script block above. What is the ghost's starting position?
 - A. (X:100, Y:50)
 - B. (X:50, Y:50)
 - C. (X:100, Y:100)
 - D. (X:0, Y:0)
 - E. (X:50, Y:100)
10. After the script block has been run, where will the ghost end up?
 - A. (X:100, Y:50)
 - B. (X:50, Y:50)
 - C. (X:50, Y:100)
 - D. (X:100, Y:100)
 - E. (X: 0, Y:0)
11. These blocks turn their sprite the specified amount of degrees clockwise or counter-clockwise (depending on which block is used); this changes the direction the sprite is facing.

All answers should be written on the ANSWER SHEET.

- A.  ,  B. 
- C.  D. 
- E. None of the above

12. Look at the Scratch stage below. Where on the stage is position (x:240, Y:180)?



- A. Top Left Corner
B. Bottom Right corner
C. The top middle
D. Top Right Corner
E. Centre of the screen

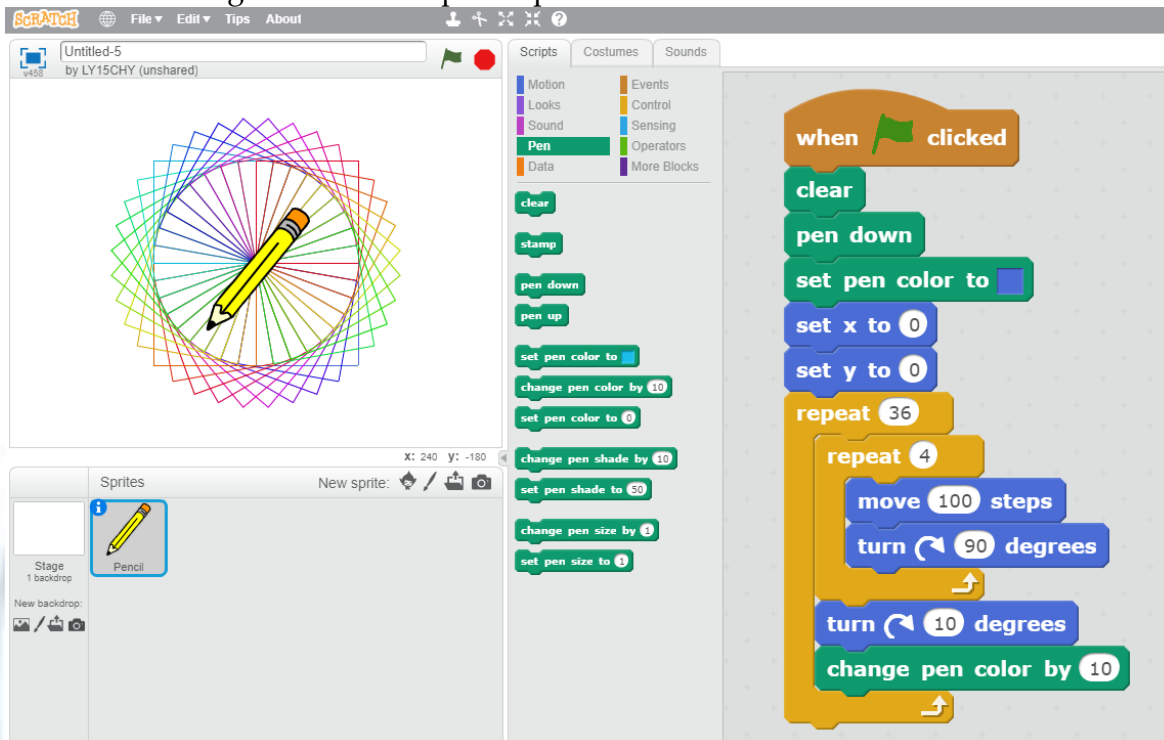
13. What happens to the shark when the spacebar is pressed AND it is touching Fish1?



All answers should be written on the ANSWER SHEET.

- A. The shark switches to costume shark-b and broadcasts a message.
- B. The shark broadcasts a message and then switches to costume shark-a
- C. The shark switches to costume shark-a
- D. The shark switches to costume shark-b
- E. Nothing happens

Refer to the following scene and script for question 14 and 15.



- 14. Look carefully at the script above. How many times is the outer repeat done?
 - A. 36
 - B. 4
 - C. 5
 - D. 6
 - E. 10
- 15. Each time the outer repeat is done, the inner repeat is done 4 times. How many times is the inner repeat done?
 - A. 4
 - B. 6
 - C. 36×4
 - D. 36
 - E. 100
- 16. What would be the answer of this mathematical expression?



- A. 10
- B. 0
- C. 5
- D. 20
- E. Infinity